Protein Engineering

Module: SC3282

By

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Lecture 03:

Protein composition and structure.

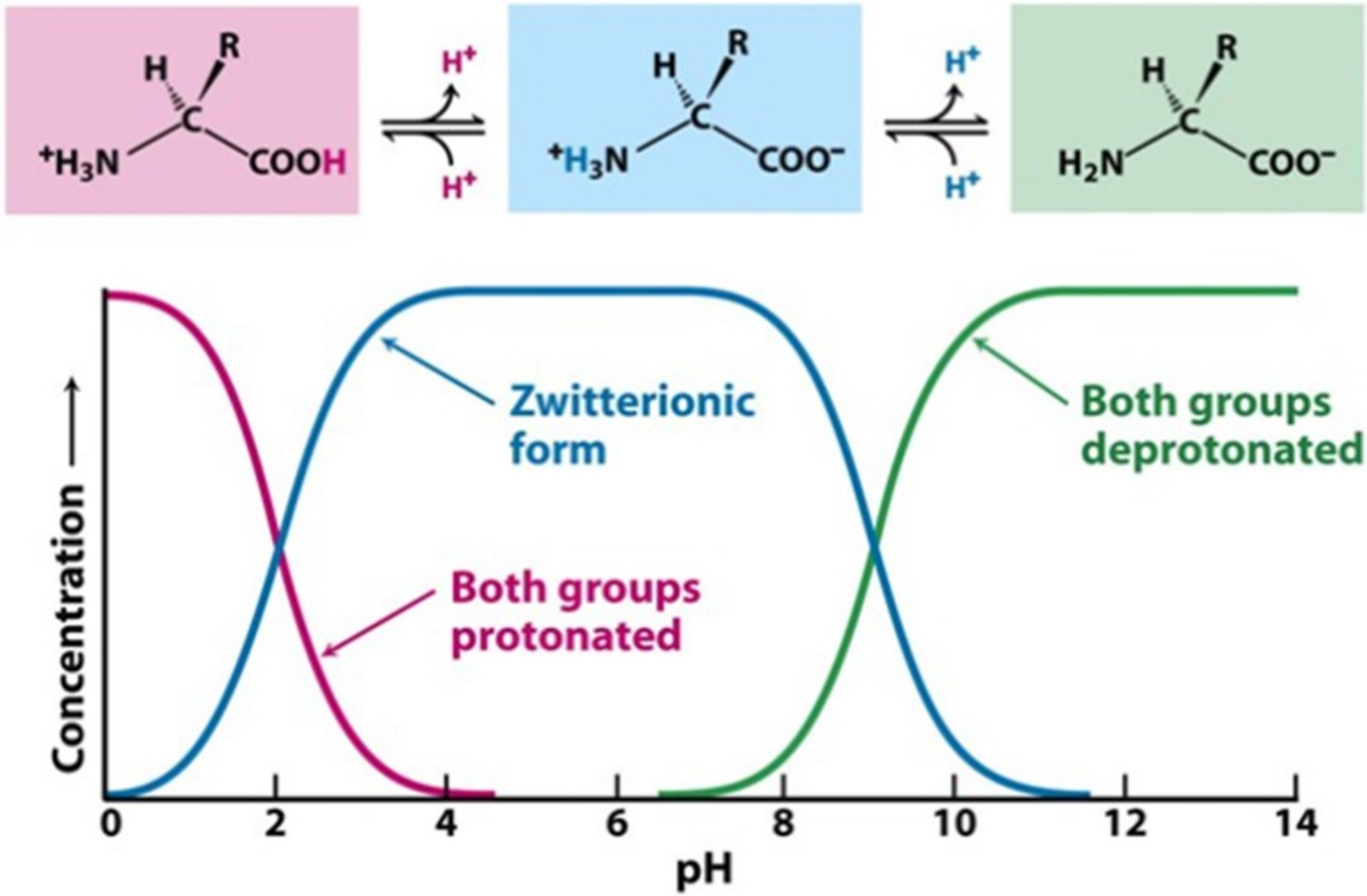
Quiz:

- If ribonuclease A is denatured using urea alone (without a reducing agent), predict the outcome on protein structure and enzymatic activity. Justify your answer

Ionization of amino acids

- A zwitterion is a functional group molecule in which at least one has a positive electrical charge and one a negative electrical charge (Dipolar ion).

- The whole molecule's net charge is negative. The best-known examples of Zwitterions are amino acids.



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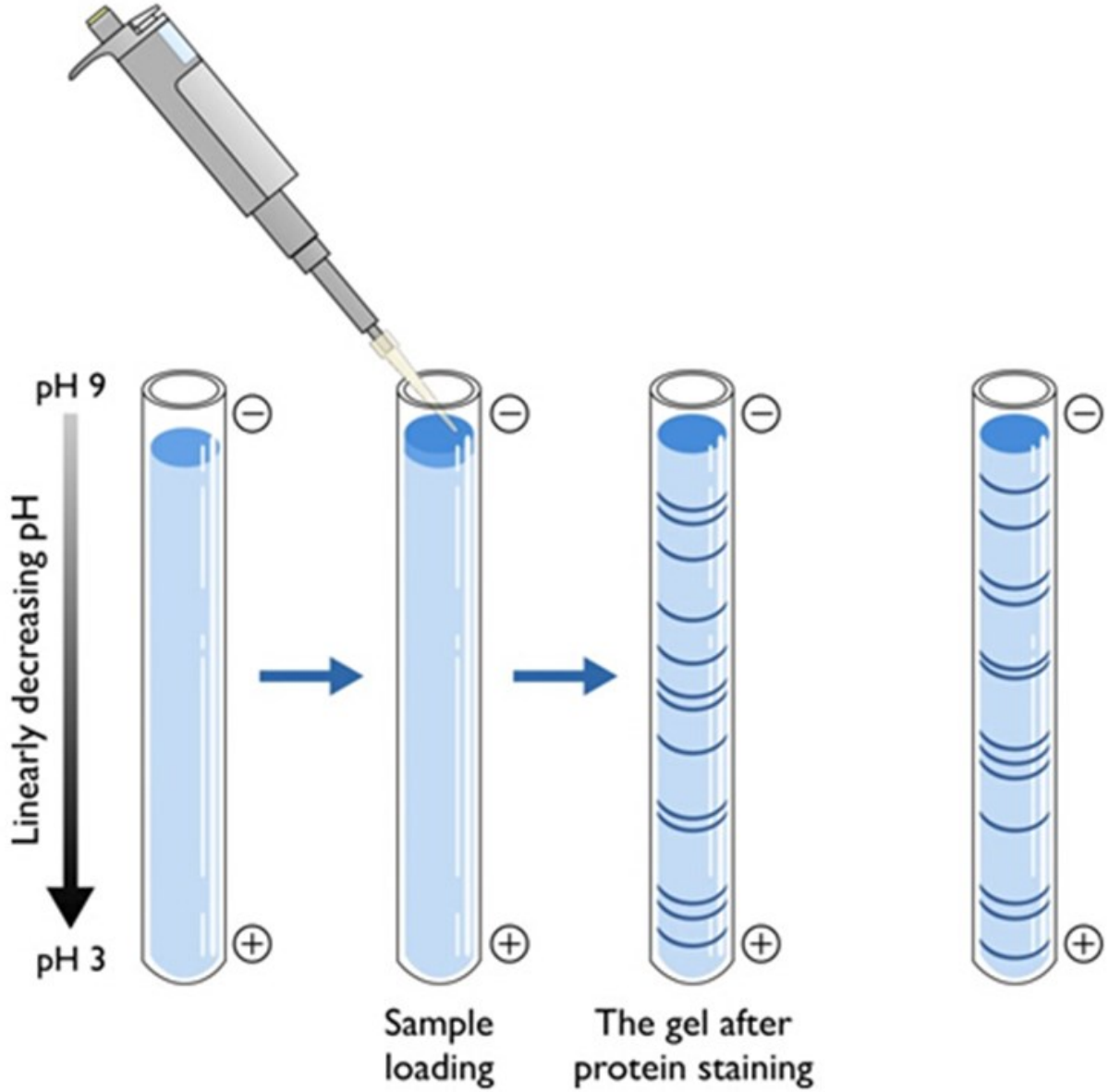
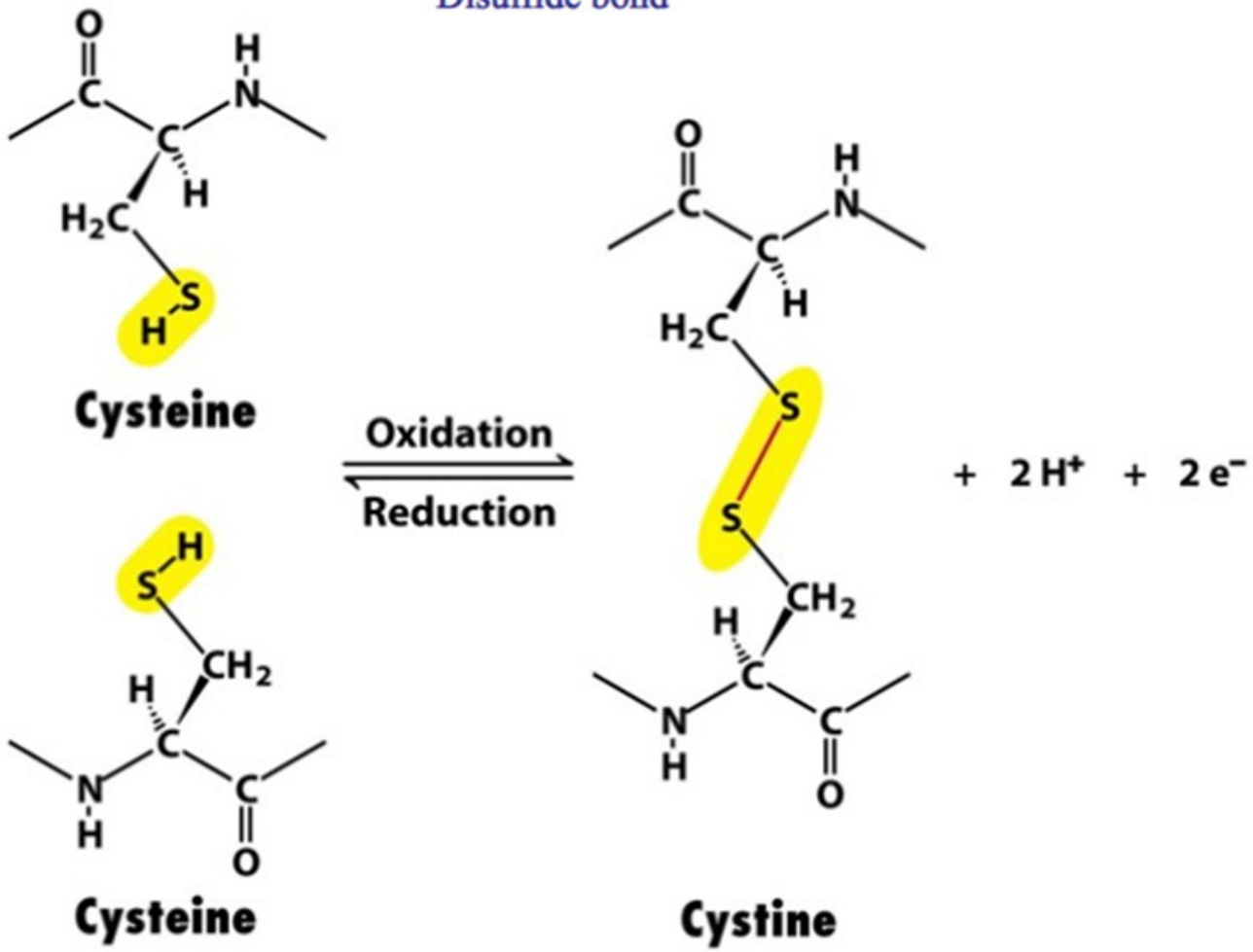
Capillary isoelectric focusing (CIEF) Technology

- Capillary isoelectric focusing allows the separation of protein/peptide mixtures, protein glycoforms and other charged variants, based on their isoelectric point (pI).

- A voltage is applied to the anolyte and catholyte tanks, establishing a pH gradient through the capillary. Proteins that are applied in the field can be separated based on their net charge, and become immobilized when they reach a pH, which equals to their pI.


pH 9
Linearly decreasing pH
pH 3
Sample loading
The gel after protein staining

Disulfide bond



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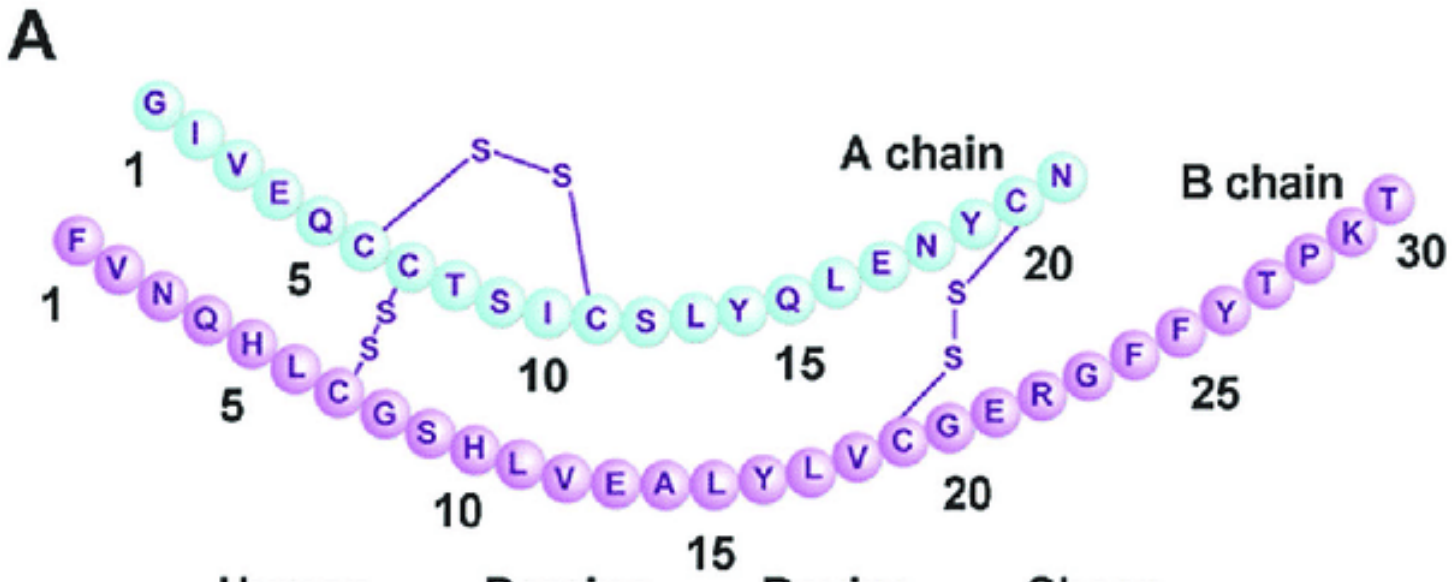
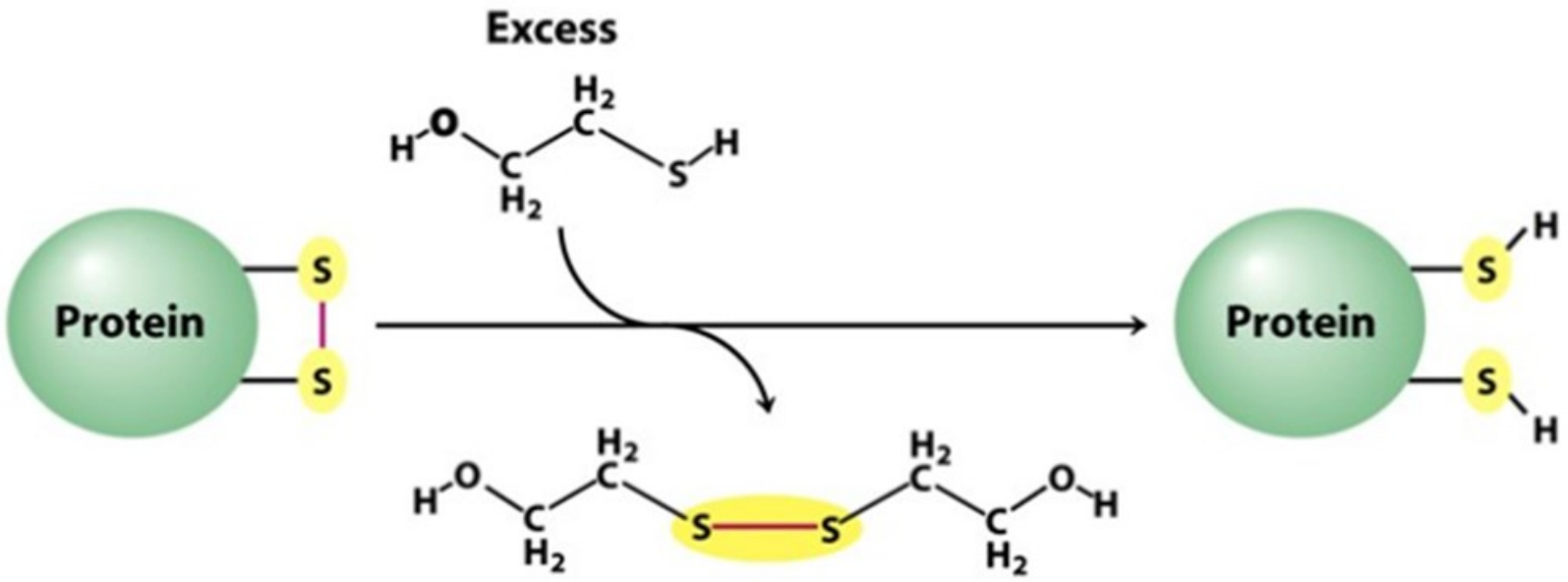
Less disulfide bonds in intracellular proteins,
Reason to add DTT, β-ME, etc

Amino acid sequence of human, porcine, bovine and sheep insulin.



	Human	Porcine	Bovine	Sheep
A8	T	T	A	A
A9	S	S	S	G
A10	I	I	V	V
B30	T	A	A	A

Reducing agent (beta-mercaptoethanol): S-S to SH



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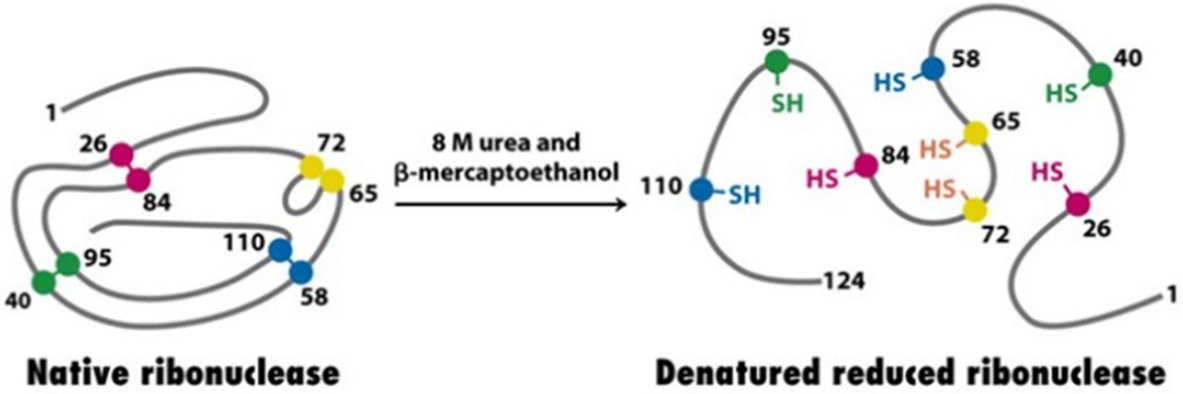

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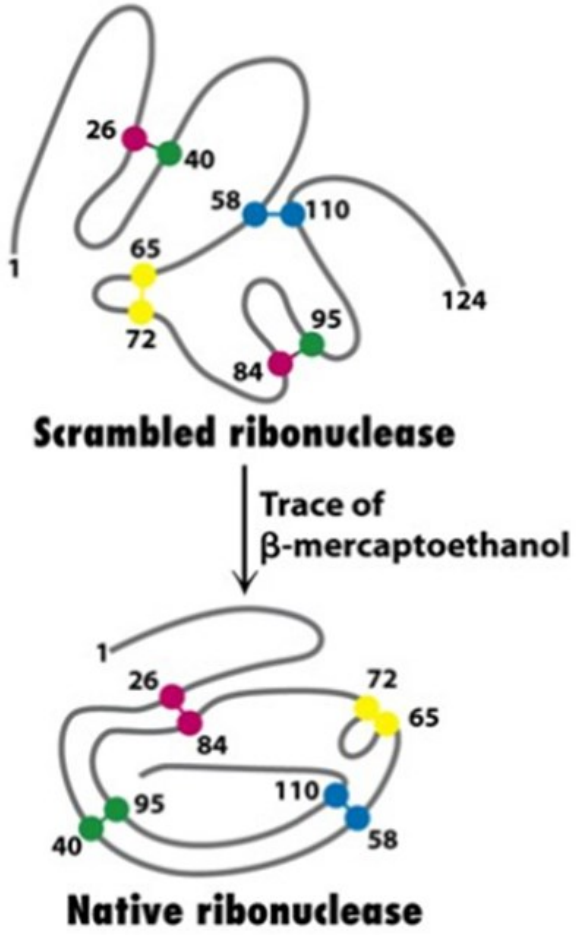
Oxidation in 8M urea
And dialysis



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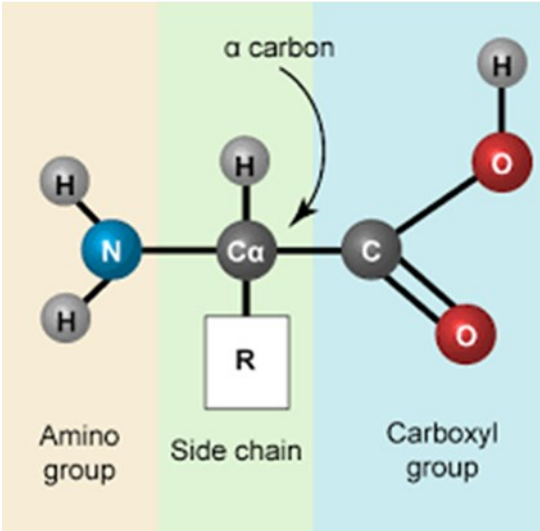
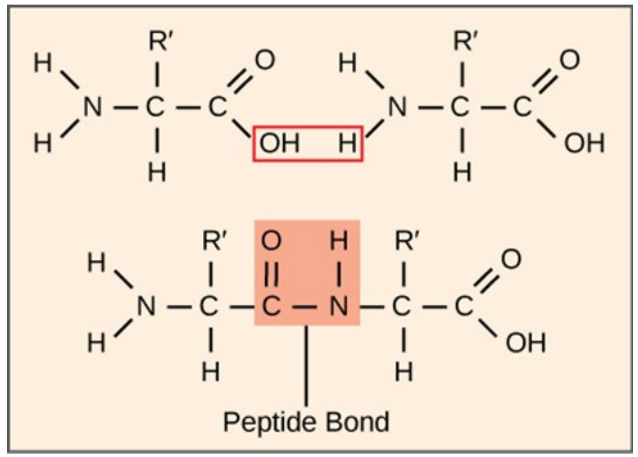
Thermodynamically preferred structure

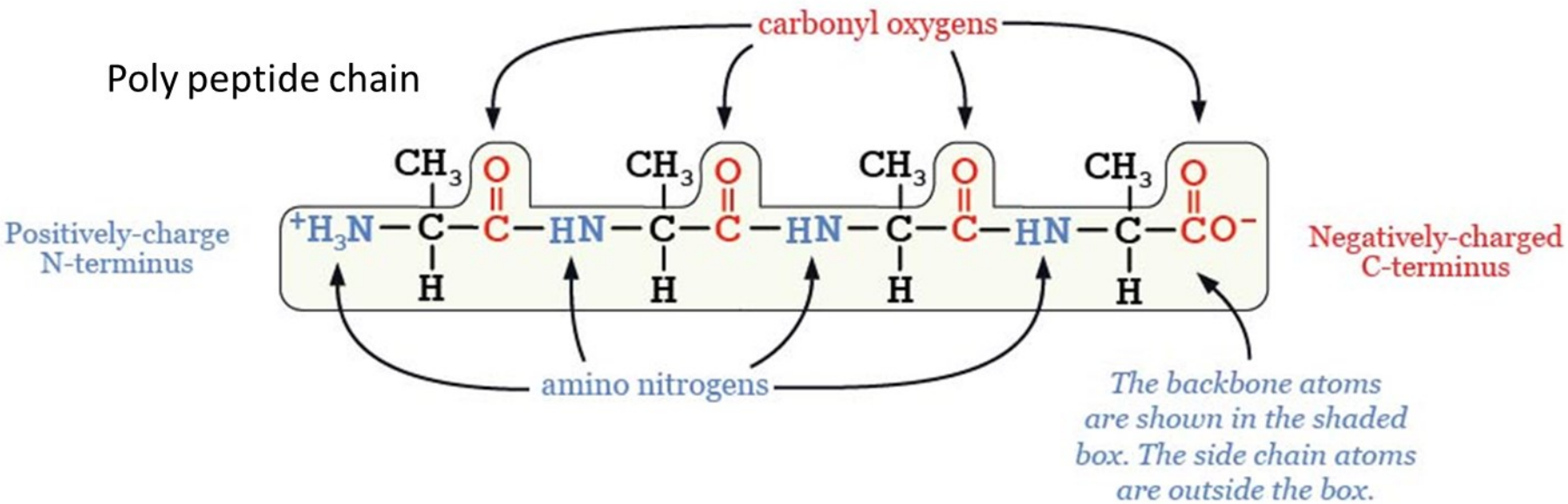
Amino acid

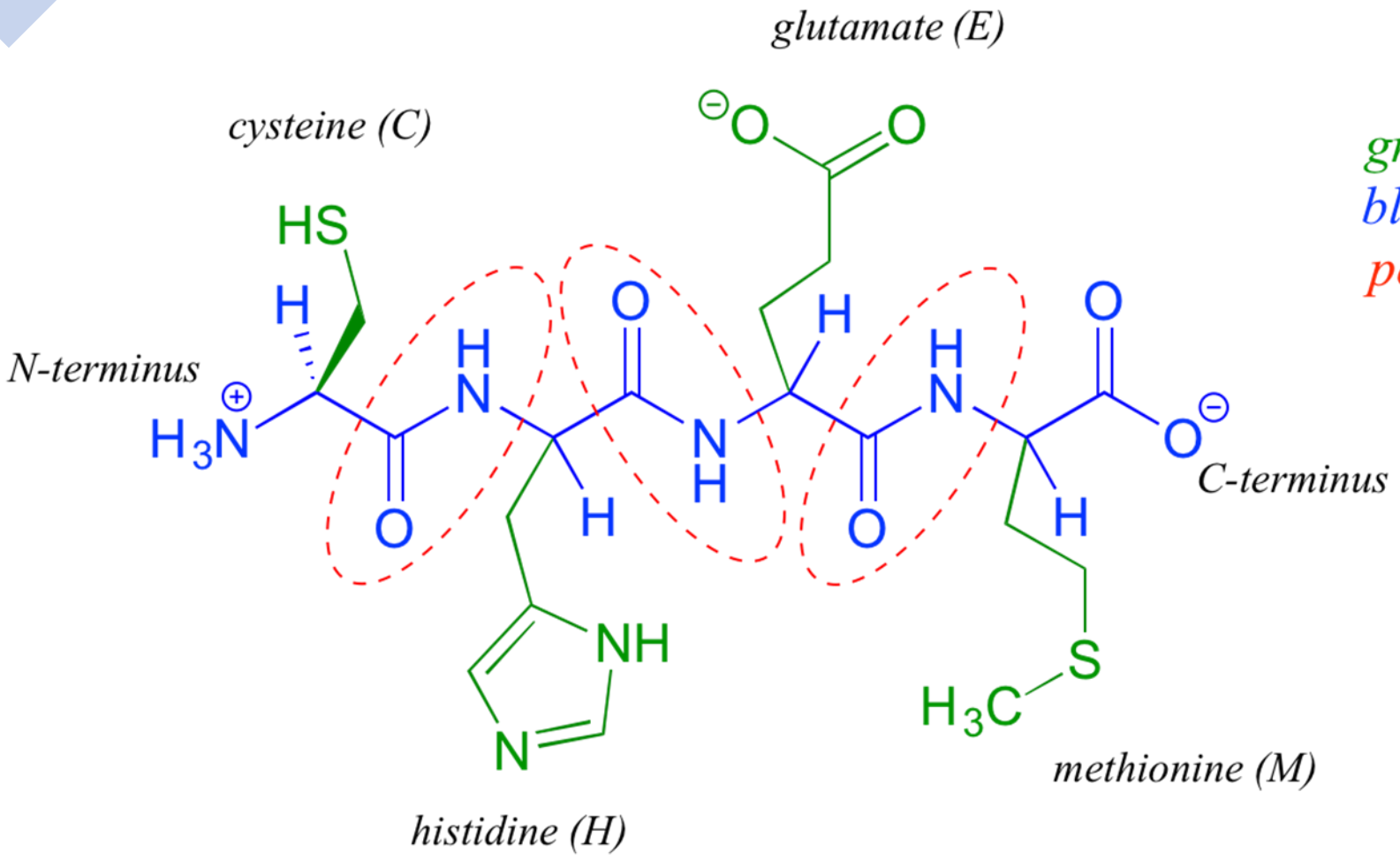


Peptide bond



Poly peptide chain



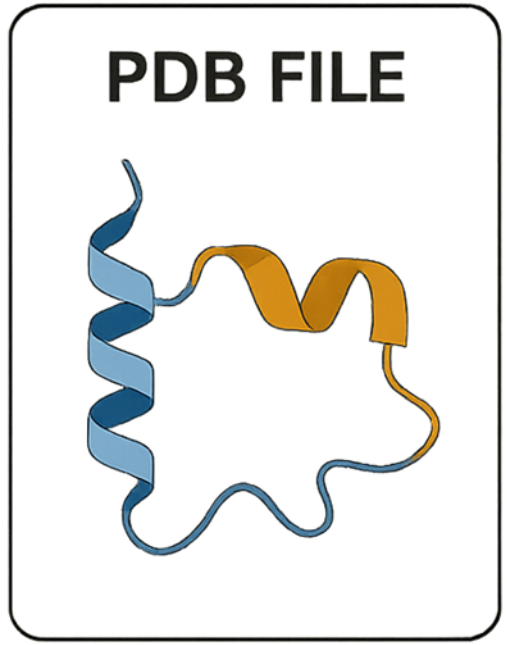


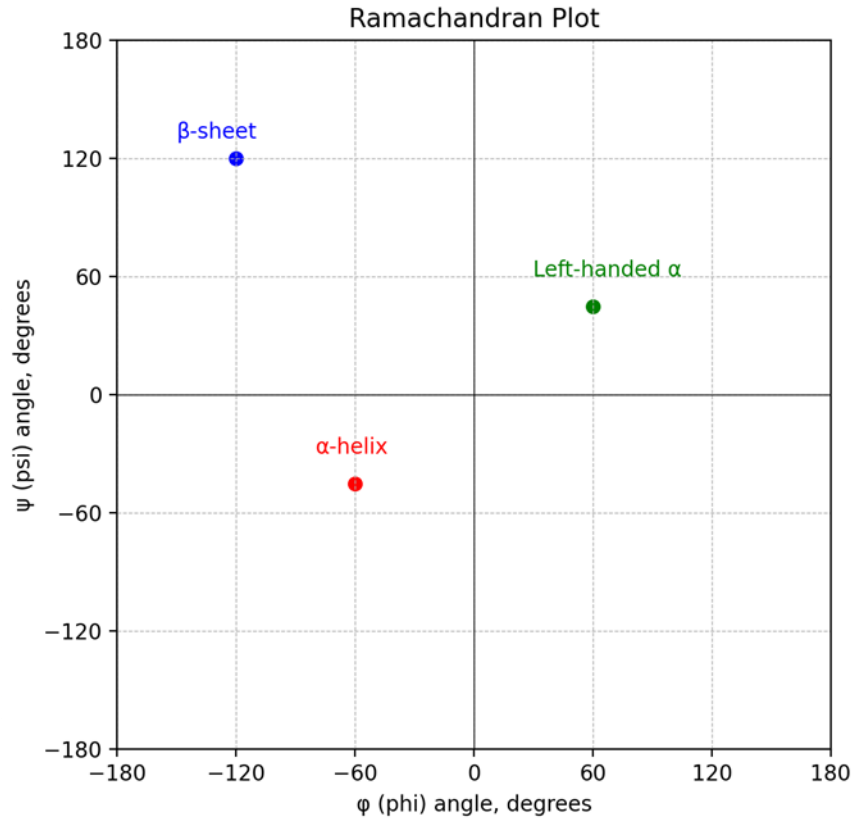
green: side chain
blue: main chain
peptide bonds circled

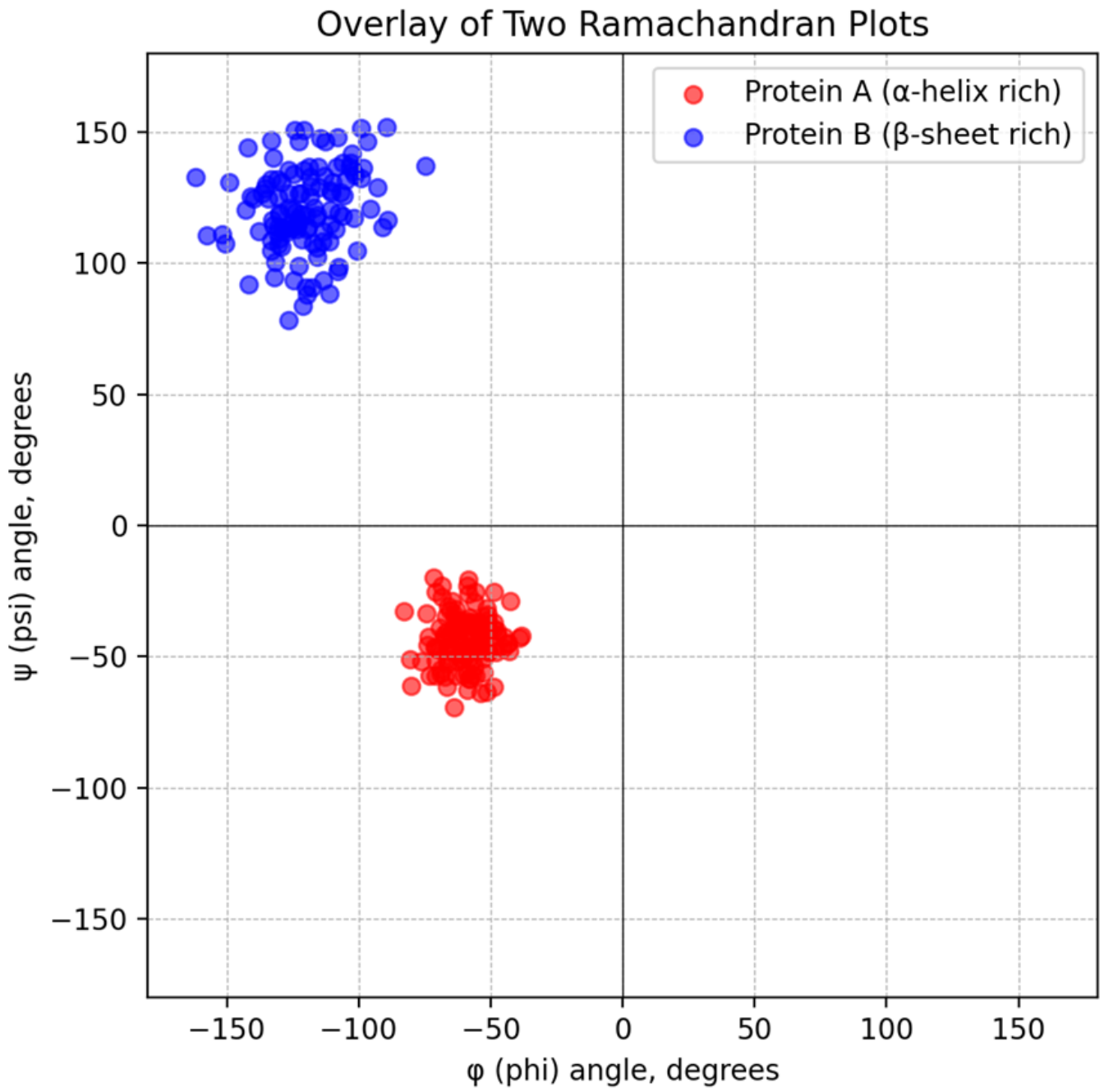
Ramachandran Plot

- The Ramachandran plot is a graphical representation of the sterically allowed and disallowed conformations of peptide backbone dihedral angles in proteins.
- It is a fundamental tool in protein structure analysis, validation and education.
- It was developed by G. N. Ramachandran and colleagues (1963) and remains essential in structural biology.

Creating a Ramachandran Plot of a Protein






Overlay of Two Ramachandran Plots
Protein A (α-helix rich)
Protein B (β-sheet rich)
150
100
50
0
−50
−100
−150
ψ (psi) angle, degrees
−150
−100
−50
0
50
100
150
φ (phi) angle, degrees

Ramachandran Plot

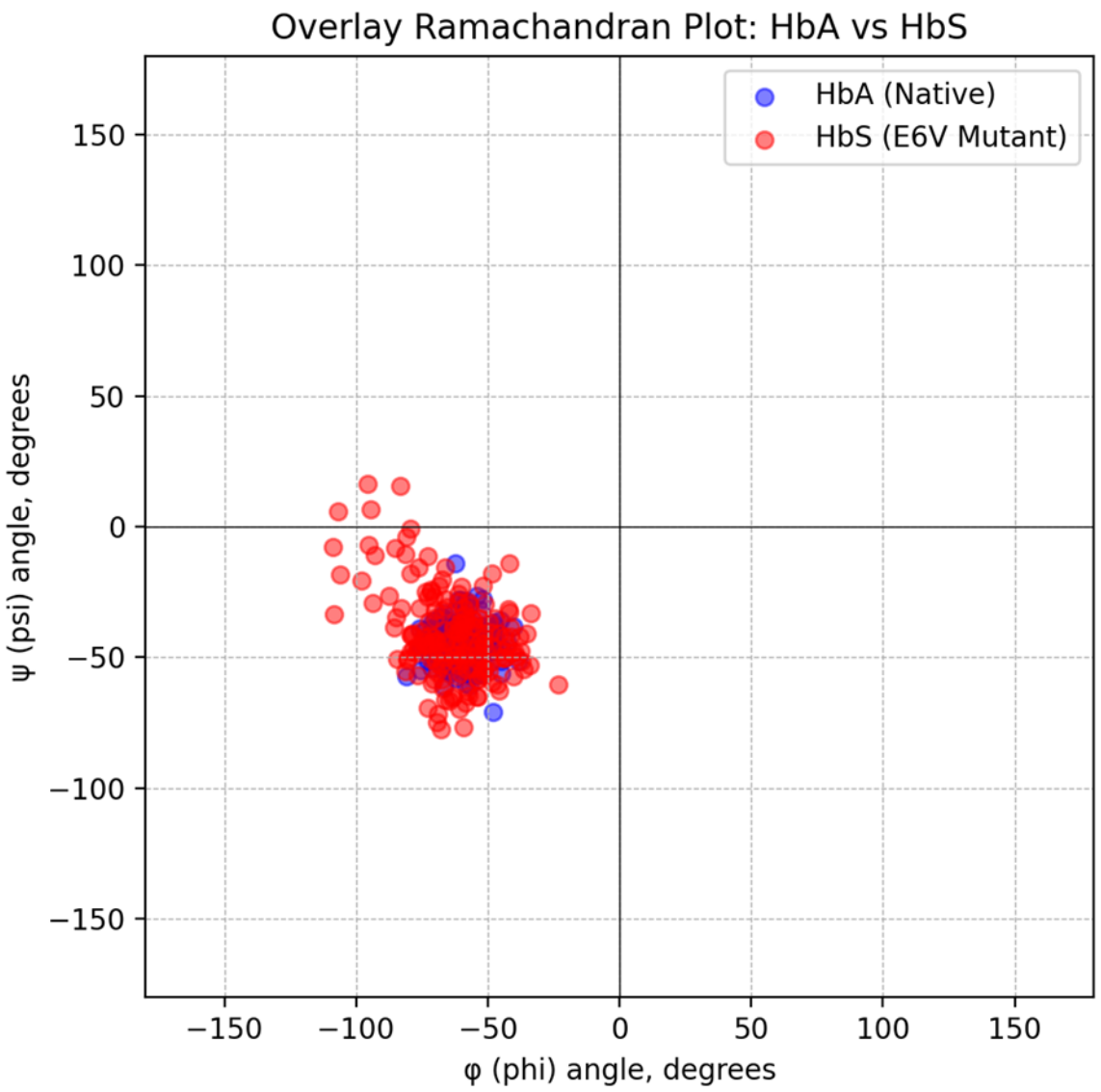
- A Ramachandran plot is a powerful method to compare the native (wild-type) and mutant forms of a protein by analyzing differences in backbone dihedral angles (ϕ and ψ).

- This comparison reveals how a mutation affects protein folding, stability, and structural integrity.

Overlay Ramachandran plot comparing native hemoglobin (HbA) and sickle-cell hemoglobin (HbS, E6V mutant).

- Blue points → HbA (native hemoglobin)
- Red points → HbS (E6V mutant hemoglobin)

- Both are largely α-helical proteins, so most points cluster near:$\phi \approx -60°$ $\psi \approx -45°$



- Overlay Ramachandran plots of HbA and HbS show that although both proteins remain largely α-helical, the HbS mutant exhibits a broader ϕ–ψ distribution and localized strain compared to the native protein.

- These subtle backbone conformational changes contribute to abnormal aggregation and disease.

Key differences visible in the overlay

1. Native HbA
   - Tight, compact clustering in the α-helix region,
   - Very few strained conformations
   - Indicates a stable and optimally folded protein

2. Mutant HbS (Glu6 → Val6)
   - Broader spread of ϕ–ψ angles
   - Appearance of residues in less favorable regions
   - Reflects local backbone strain induced by the hydrophobic valine
   - These subtle changes facilitate: Abnormal intermolecular interactions, Polymerization of HbS under low oxygen conditions

Why this example is biologically important

- The E6V mutation does not drastically unfold hemoglobin
- Ramachandran analysis reveals small but significant backbone distortions
- These changes are sufficient to cause: Fiber formation, Red blood cell sickling, Sickle-cell disease pathology

- ✅ This demonstrates how even a single amino-acid substitution can be detected and interpreted through Ramachandran plots.

1D gel:

https://www.youtube.com/watch?v=S7NIkBy38To

2D gel:

https://www.youtube.com/watch?v=FHvblg338Gg

Two broad categories of proteins according to its composition, they are:

Protein Classification (*Based on Composition*)

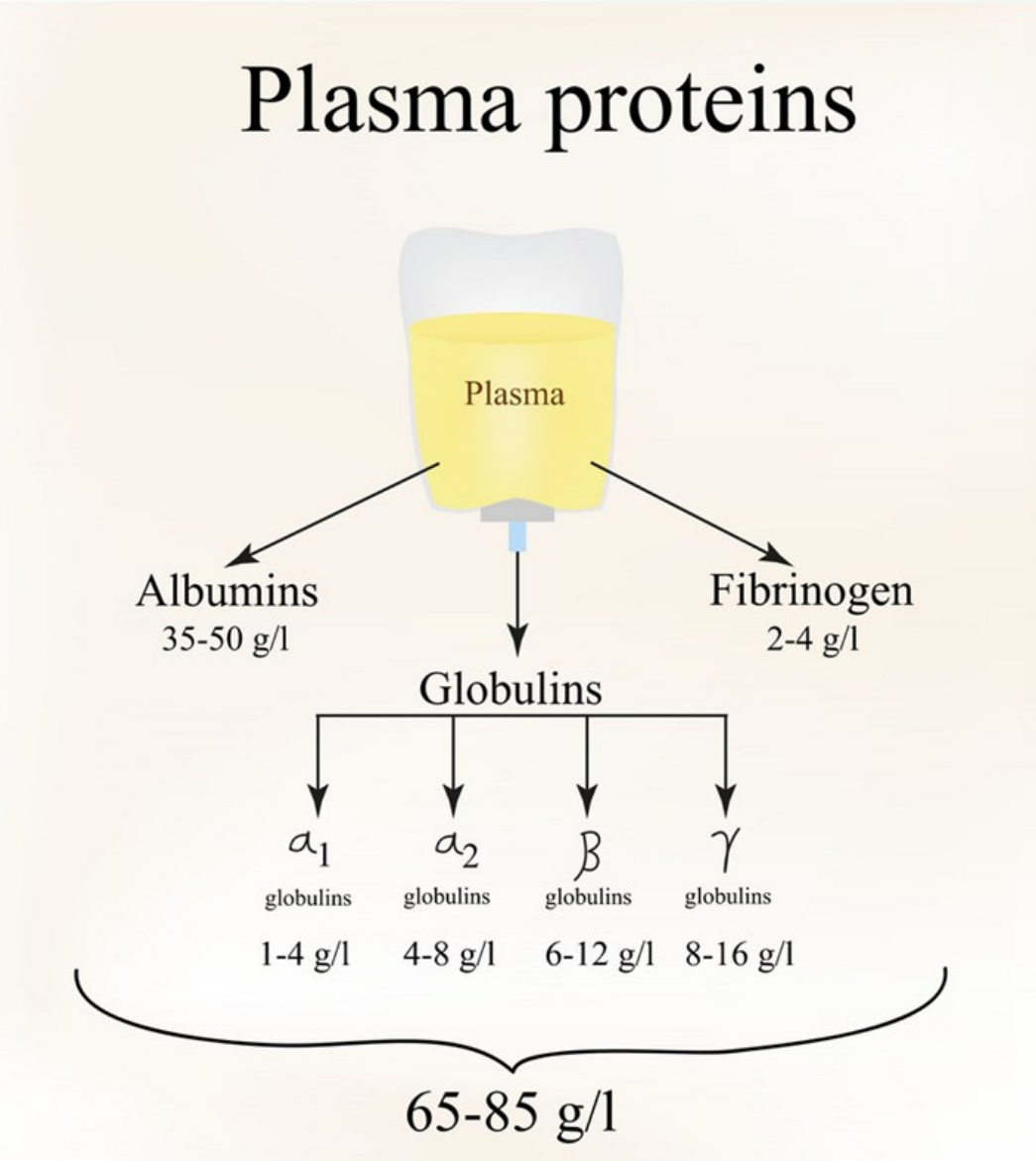
- **Simple Protein**
- **Conjugated Protein**
  - *Phosphoproteins*
  - *Glycoproteins*
  - *Nucleoproteins*
  - *Chromoproteins*
  - *Lipoproteins*
  - *Flavoproteins*
  - *Metalloproteins*

Simple proteins

- Simple proteins composed of ONLY amino acids.
- Proteins may be fibrous or globular.
- They possess relatively simple structural organization.
- Example: Collagen, Myosin, Insulin, Keratin

Where are simple proteins found?

- These are mainly found in blood plasma, gastric and salivary mucin, immunoglobulins, human chorionic gonadotropins, etc.



Globulins

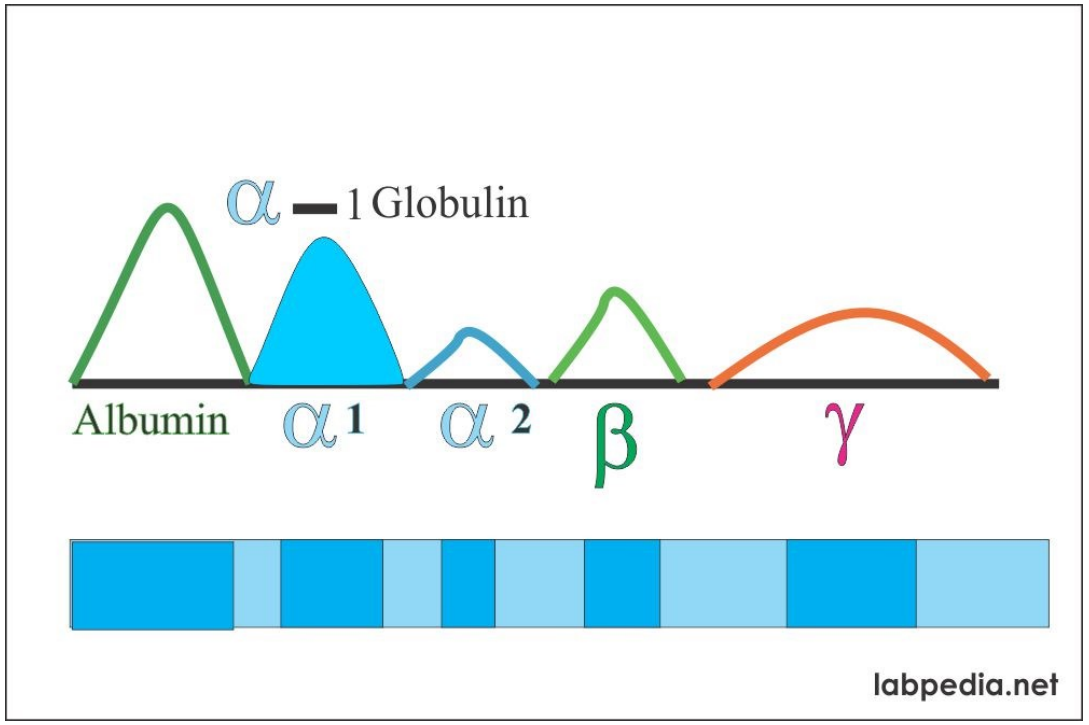
α_1-globulins – transport of lipids, thyroxin, corticosteroid hormones;

α_2-globulins - transport of lipids, copper ions;

β-globulins - transport of lipids, iron and protective (participation of β-globulins in immune reactions as antitoxins);

γ-globulins as immunoglobulins functions.

Protein Serum Electrophoresis (Total protein, albumin and globulin)



Albumin

Open flexible structure

- Higher solubility
- Higher tyrosine fluorescence intensity
- Higher foam capacity
- Higher content of unordered secondary structure fraction
- Lower content of aromatic and hydrophobic amino acids

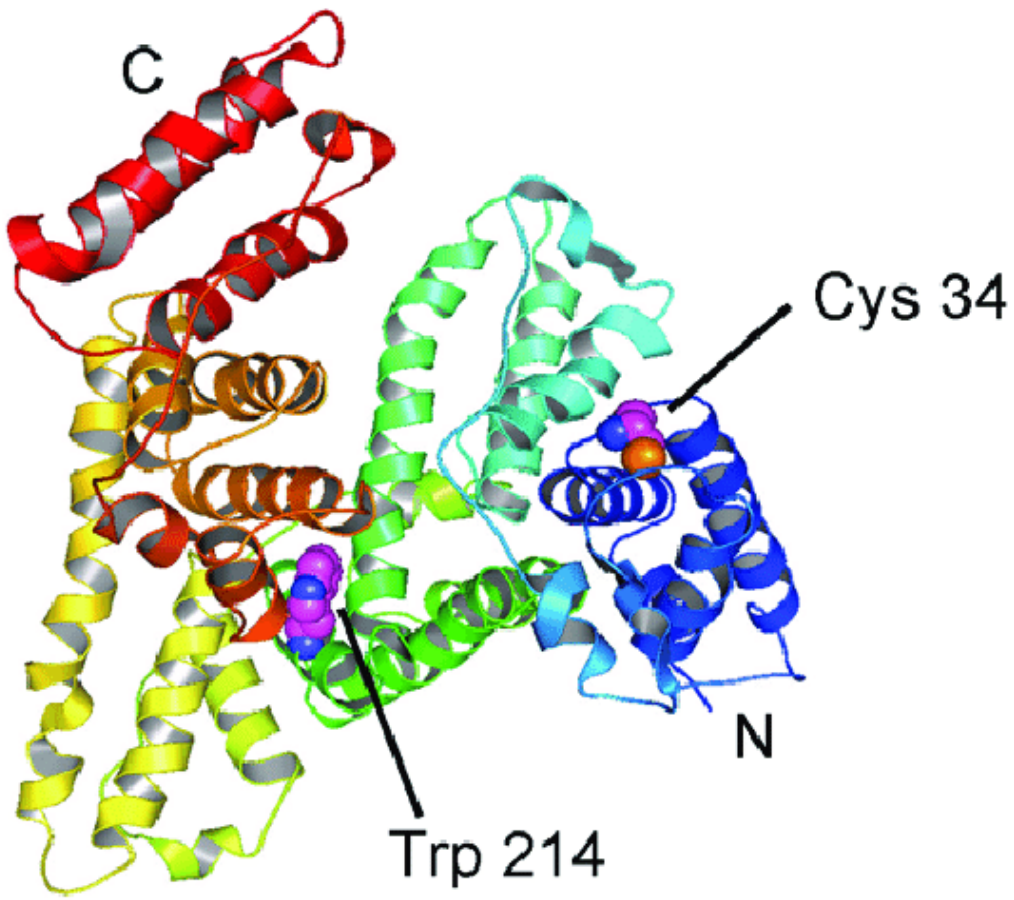
Globulin

Closed non-flexible structure

- Lower solubility
- Lower tyrosine fluorescence intensity
- Lower foam capacity
- Lower content of unordered secondary structure fraction
- Higher content of aromatic and hydrophobic amino acids

Albumin

- It is a protein made by your liver. Albumin helps keep fluid in your blood stream, so it doesn't leak into other tissues. It is also carring various substances throughout your body, including hormones, vitamins, and enzymes. Low albumin levels can indicate a problem with your liver or kidneys.



Conjugated proteins

- Conjugated proteins are complex proteins.
- They contain one or more non-amino acid components.
- Here the protein part is tightly or loosely bound to one or more non-protein part(s).
- The non-protein parts of these proteins are called prosthetic groups.
- The prosthetic group may be metal ions, carbohydrates, lipids, phosphoric acids, nucleic acids and FAD.
- The prosthetic group is essential for the biological functions of these proteins.
- Conjugated proteins are usually globular in shape and are soluble in water.
- Most of the enzymes are conjugated proteins.

Conjugated proteins

Conjugated Protein

- *Phosphoproteins*
- *Glycoproteins*
- *Nucleoproteins*
- *Chromoproteins*
- *Lipoproteins*
- *Flavoproteins*
- *Metalloproteins*

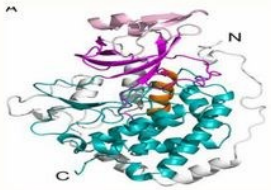
Conjugated proteins are further classified as follows:

- **Phosphoprotein:** Prosthetic group is phosphoric acid, Example- Casein of milk, Vitellin of egg yolk.
- **Glycoproteins:** Prosthetic group is carbohydrates, Example – Most of the membrane proteins, Mucin (component of saliva).
- **Nucleoprotein:** Prosthetic group is nucleic acid, Example – proteins in chromosomes, structural proteins of ribosome.
- **Chromoproteins:** Prosthetic group is pigment or chrome, Example: Haemoglobin, Phytochrome and Cytochrome.
- **Lipoproteins:** Prosthetic group is Lipids, Example: Membrane proteins
- **Flavoproteins:** Prosthetic group is FAD (Flavin Adenine Dinucleotide), Example: Proteins of Electron Transport System (ETS).
- **Metalloproteins:** Prosthetic group is Metal ions, Example: Nitrate Reductase.

- **Phosphoproteins** offer a unique minute-by-minute record of ongoing signal pathway events of high functional relevance to therapeutic target selection and the prediction of toxicity.

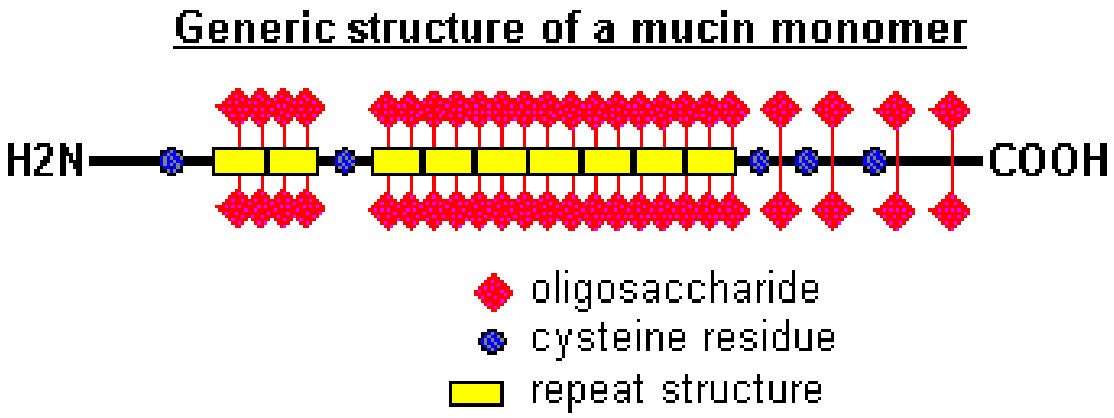
- Phosphorylation and dephosphorylation of structural and regulatory proteins are major intracellular control mechanisms

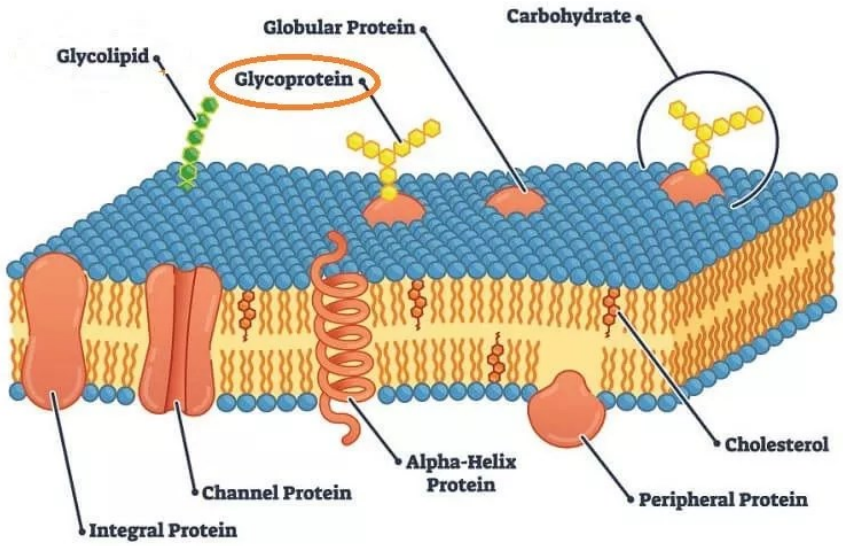
Structure of casein



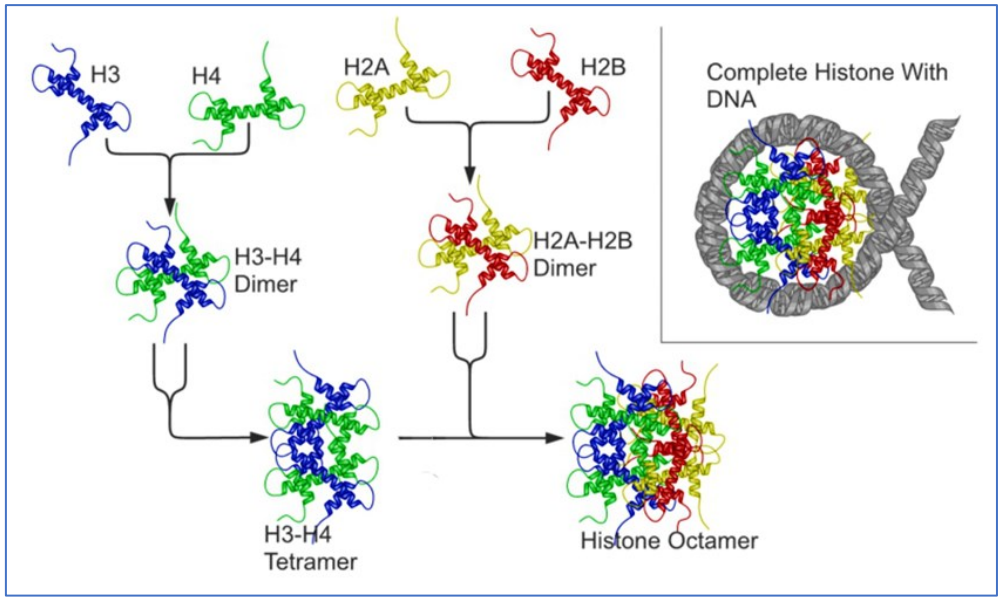
- Is a phosphoprotein which has **phosphate groups** attached to some of the amino acid side chains.
- Mostly these amino acid are **Serine** and **Threonine**.
- Casein is a mixture of at least three similar proteins, which differ primarily in molecular weight and amount of phosphorus they contain (number of phosphate groups).
- Casein is made up of the main 3 types of proteins: **α-casein**, **β-casein**, and **κ-casein**.
- All casein proteins have different hydrophobic and hydrophilic regions along the protein chain.

- **Glycoproteins** are molecules that comprise protein and carbohydrate chains that are involved in many physiological functions including immunity. Many viruses have glycoproteins that help them enter bodily cells, but can also serve to be important therapeutic or preventative targets.

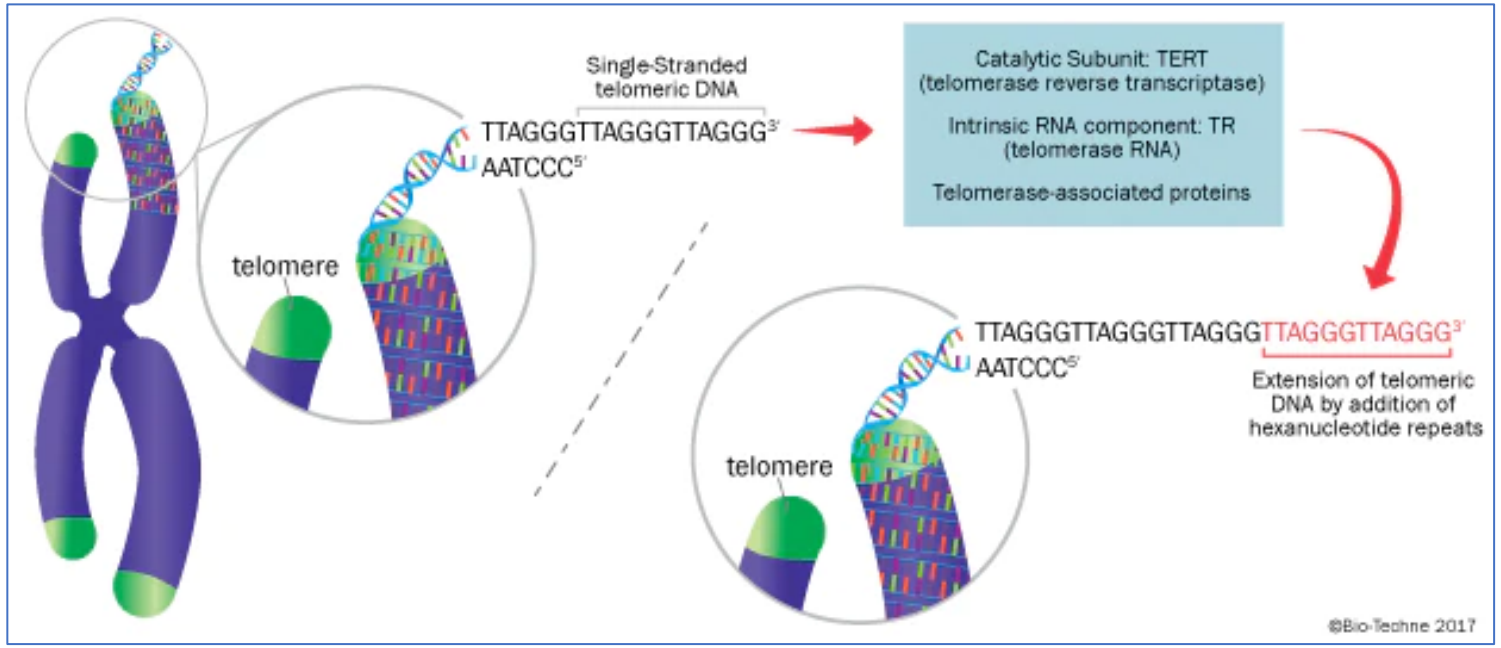




- **Nucleoprotein**, conjugated protein consisting of a protein linked to a nucleic acid, either DNA (deoxyribonucleic acid) or RNA (ribonucleic acid). The protein combined with DNA is commonly either histone or protamine; the resulting nucleoproteins are found in chromosomes.



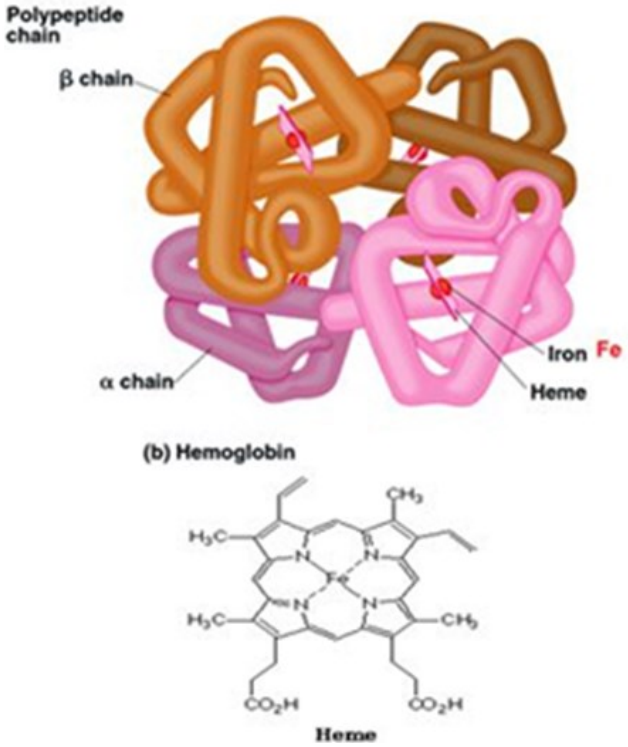
- **Telomeres** are nucleoprotein structures at the ends of eukaryotic chromosomes that protect linear chromosomes against damage by endogenous nucleases and erroneous recognition as unrepaired chromosomal breaks. ... The telomeric cap proteins of diverse organisms are less conserved than one might expect.

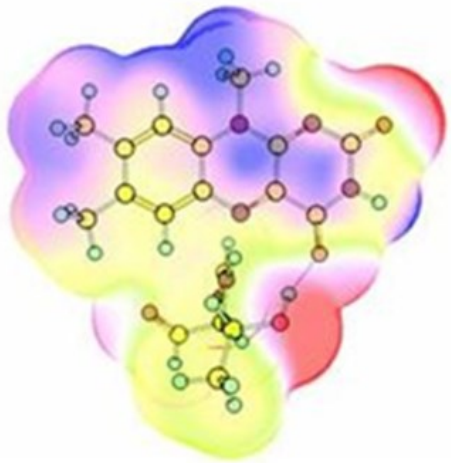


- **Chromoprotein** is a conjugated protein that contains a pigmented prosthetic group (or cofactor).

- A common example is Hemoglobin, which contains a heme cofactor, which is the iron-containing molecule that makes oxygenated blood appear red.

- Other examples of chromoproteins include cytochromes, phytochromes and flavoproteins.



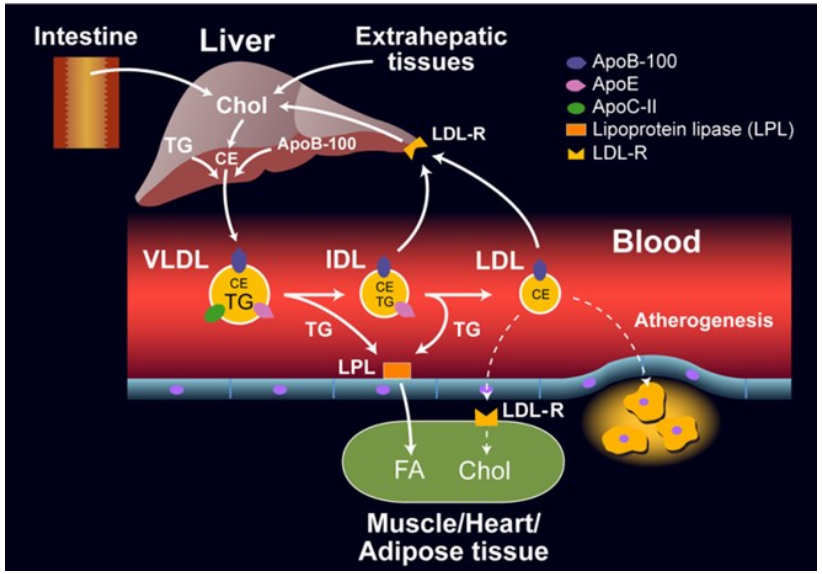


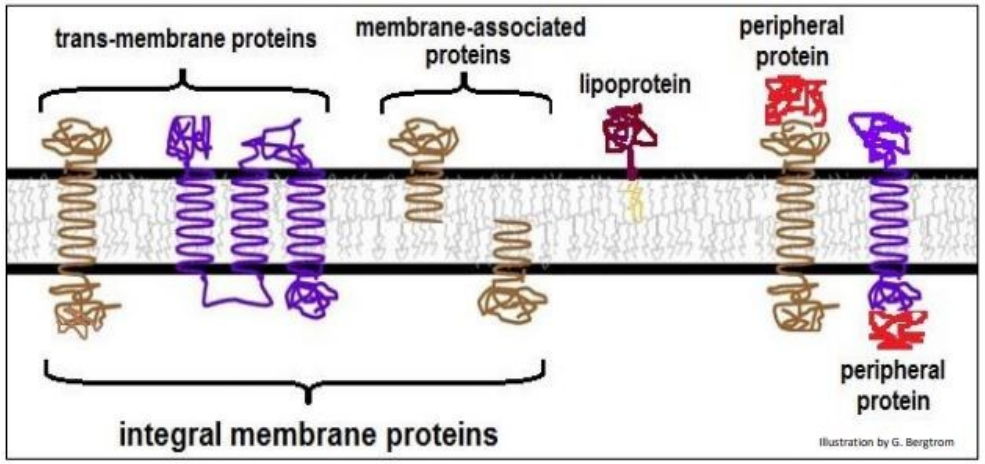
Cytochromes P450

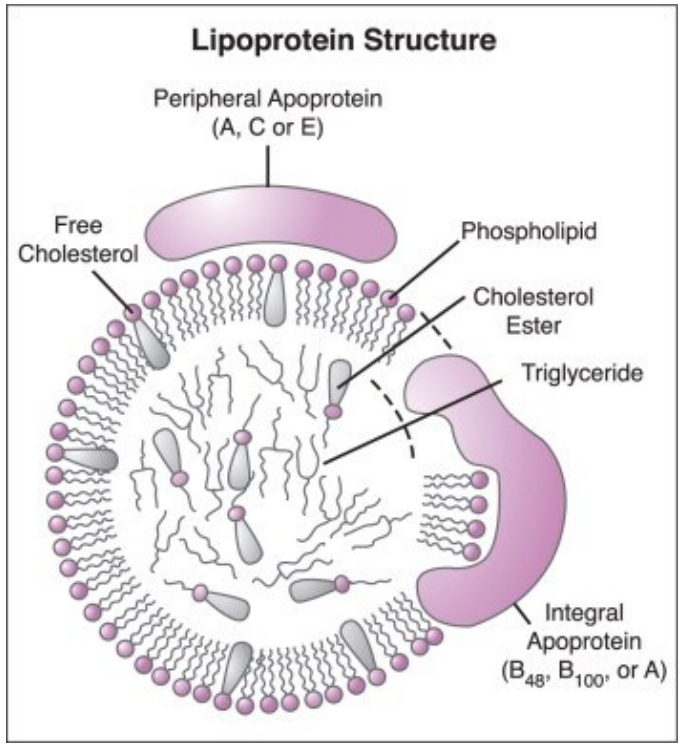
- **Lipoproteins** are substances made of protein and fat that carry cholesterol through your bloodstream.

- There are two main types of cholesterol: High-density lipoprotein (HDL), or "good" cholesterol. Low-density lipoprotein (LDL), or "bad" cholesterol.

- A lipoprotein is a biochemical assembly whose primary function is to transport hydrophobic lipid molecules in water, as in blood plasma or other extracellular fluids.





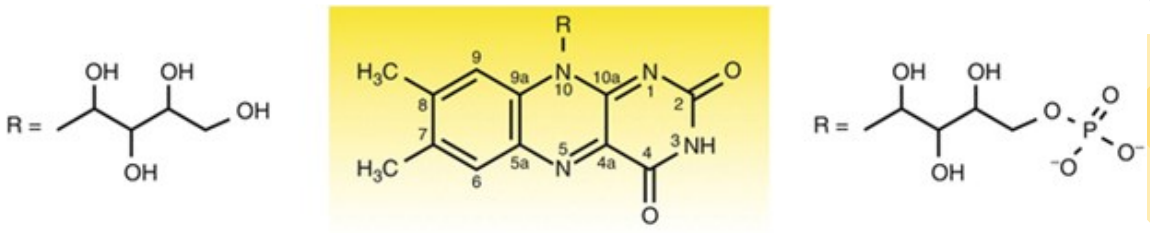


- **Flavoproteins** are proteins that contain a nucleic acid derivative of riboflavin: the flavin adenine dinucleotide or flavin mononucleotide. Flavoproteins are involved in a wide array of biological processes, including removal of radicals contributing to oxidative stress, photosynthesis, and DNA repair.

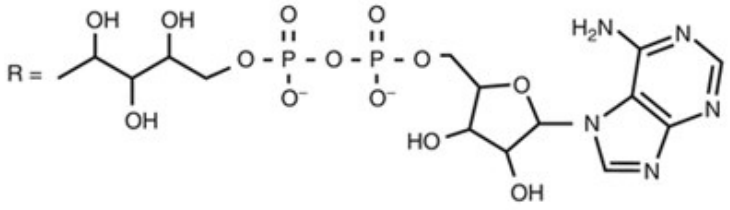
- Flavoproteins have either FMN or FAD as a prosthetic group or as a cofactor.

- About 5-10% of flavoproteins have a covalently linked FAD.

Flavins



riboflavin (vitamin B2) | 7,8-dimethylisoalloxazine | flavin mononucleotide (FMN)



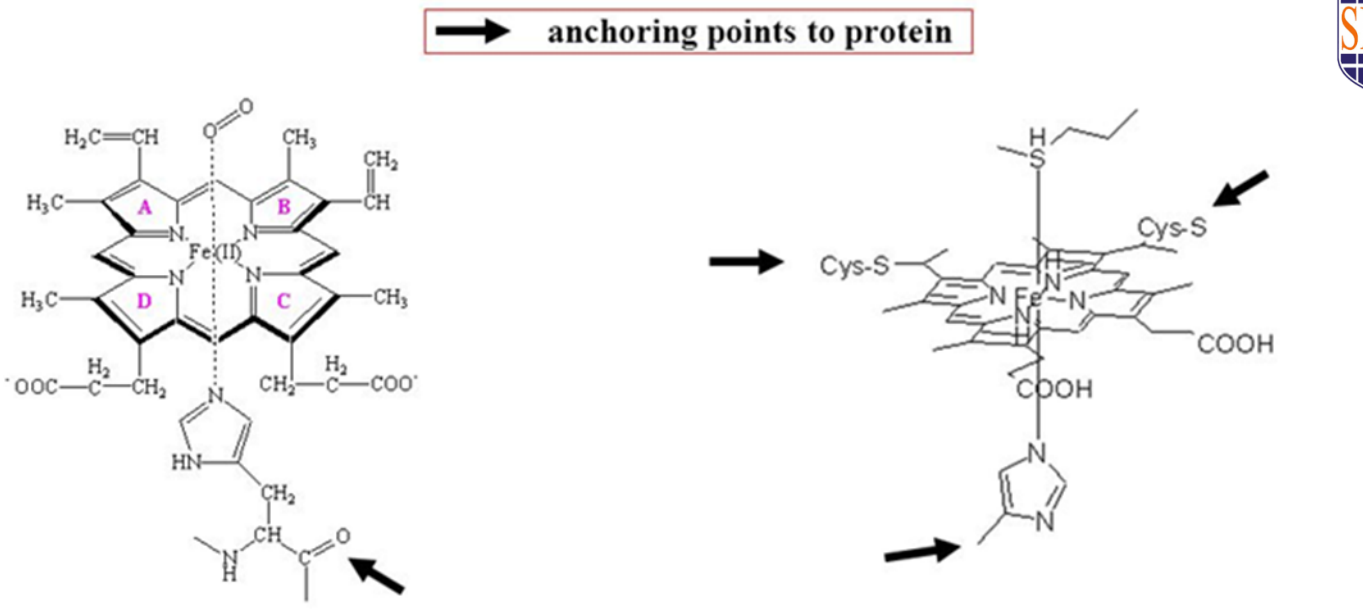
flavin adenine dinucleotide (FAD)

Flavoprotein

- **Metalloprotein** is a generic term for a protein that contains a metal ion cofactor. A large proportion of all proteins are part of this category.

- For instance, at least 1000 human proteins (out of ~20,000) contain zinc-binding protein domains although there may be up to 3000 human zinc metalloproteins.

- Metalloproteins represent some of the best inorganic catalysts known to mankind, catalyzing with high efficiency some of the most important biological reactions in nature, from photosynthesis, respiration, water oxidation to signal transduction, and complex chemical transformations.



Hemoglobin/myoglobin

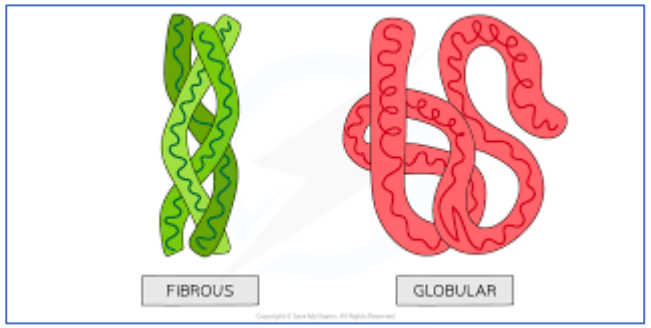
Cytochrome c (involved in respiratory chain)

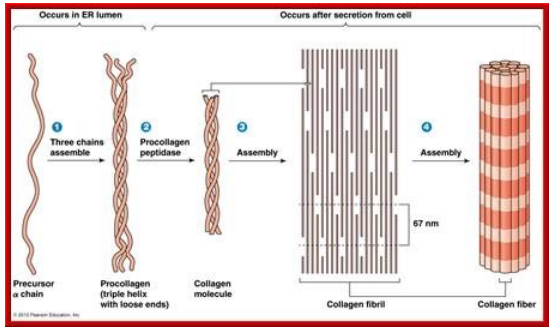
Zinc protein	Biological function
Alcohol dehydrogenase (ADH)	Oxidoreductase, alcohol breakdown
Farnesyl transferase (FT)	Transferase, signal transduction
Carboxypeptidase A (CPA)	Hydrolase, peptide cleavage
Carbonic anhydrase (CA)	Lyase, activation of small molecules
Phosphomannose isomerase (PMI)	Isomerase, isomerization of fructose-6-phosphate
DNA ligase III	Ligase, DNA repair
Zinc Fingers (ZF)	DNA transcription, DNA repair
Zinc $^{2+}$	Growth, neurotransmission

Protein Classification
(Based on Structure)

- Fibrous Proteins
- Globular Proteins
- Intermediate Proteins

Fibrous proteins





- They are linear (long fibrous) in shape.
- Secondary structure is the most important functional structure of fibrous proteins.
- Usually, these proteins do not have tertiary structures.
- Physically fibrous proteins are very tough and strong.
- They are insoluble in the water.
- Long parallel polypeptide chains cross linked at regular intervals.

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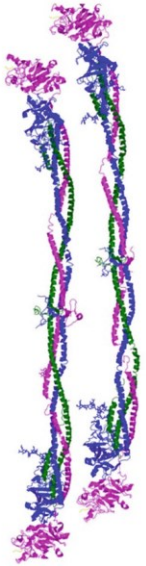
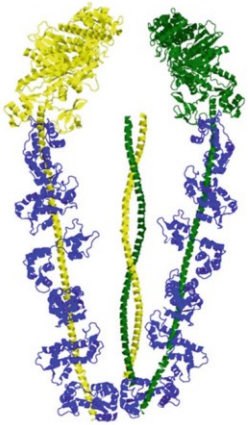
Collagen /5ctd/

Keratin /4zry/

Fibrinogen /1fza/



Myosin /2dfs/

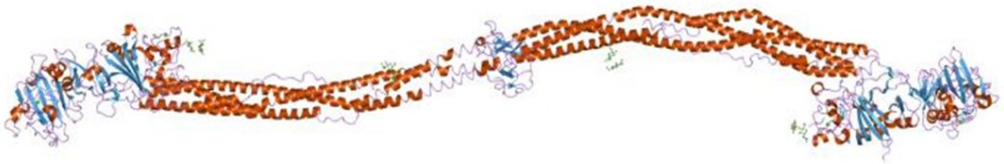


Globular Proteins

- Globular proteins are spherical or globular in shape.
- The polypeptide chain is tightly folded into spherical shapes.
- Tertiary structure is the most important functional structure in globular proteins.
- Physically they are soft than fibrous proteins.
- They are readily soluble in water.
- Most of the proteins in the cells belong to the category of globular proteins.
- Functions: Form enzymes, antibodies and some hormones.
- Example: Insulin, Haemoglobin, DNA Polymerase and RNA Polymerase

Intermediate Proteins

- Their structure is intermediate to linear and globular structures.
- They are short and more or less linear shaped proteins
- Unlike fibrous proteins, they are soluble in water.
- Function: blood clotting proteins
- Example: Fibrinogen



Tutorial:

- What are the differences between telomere and telomerase?
- Discuss the telomerase structure and clinical implications.

Thank you